

Phase 2-Configure GoldenGate Replication from Oracle Hash to Aurora Postgres

- [DMS Replication Setup \(Full-Load\)](#)

• AURORA POSTGRES HASH DATABASE SETUP

- Cluster setup Via Ips
- Database Setup
- Schema setup

DMS Replication Setup (Full-Load)

- **Configure GoldenGate on source**
- login to gg hub **ggpsphpp05.sbg-ppd-us-west-2/binaries/dataguard/scripts/jdk-8u371-linux-x64.tar.gz**

```
1 cd /u01/software
2 unzip 213000_fbo_ggs_Linux_x64_Oracle_shiphome.zip
3 mkdir -p /u01/ogg/psphpp05/oracle/21.9.0.0
4 vi /u01/software/oggcore.rsp
5 cd /u01/ogg/jdk1.8/
6 aws s3 cp s3://bits-sbg-qbo-ppd-us-west-2/binaries/dataguard/scripts/jdk-8u371-linux-x64.tar.gz
7 tar -zxvf jdk-8u371-linux-x64.tar.gz
8 cd /u01/software/fbo_ggs_Linux_x64_Oracle_shiphome/Disk1/
9 ./runInstaller -silent -nowait -responseFile /u01/software/oggcore.rsp
```

Output

```
1 [oracle@ggpsphpp05 Disk1]$ ./runInstaller -silent -nowait -responseFile /u01/software/oggcore.rsp
2 Starting Oracle Universal Installer...
3
4 Checking Temp space: must be greater than 120 MB. Actual 34267 MB Passed
5 Checking swap space: must be greater than 150 MB. Actual 1023 MB Passed
6 Preparing to launch Oracle Universal Installer from /tmp/OraInstall2023-07-10_11-50-17AM. Please wait ...
7 [oracle@ggpsphpp05 Disk1]$ You can find the log of this install session at:
8 /u01/app/oraInventory/logs/installActions2023-07-10_11-50-17AM.log
9 Successfully Setup Software.
10 The installation of Oracle GoldenGate Core was successful.
11 Please check '/u01/app/oraInventory/logs/silentInstall2023-07-10_11-50-17AM.log' for more details.
```

- Apply Patch

```
1 cd /u01/software
2 unzip p6880880_210000_Linux-x86-64.zip
3 unzip p34958358_2190000GGRU_Linux-x86-64.zip
4 updated GG_HOME in /l/orcl file
5 cd $GG_HOME
6 export ORACLE_HOME=/u01/ogg/psphpp05/oracle/21.9.0.0
7 OPatch/opatch version
8
9 mv OPatch OPatch.old
10 cp -rp /u01/software/OPatch .
11 cd /u01/software/34958358/
12 /u01/ogg/psphpp05/oracle/21.9.0.0/OPatch/opatch lsinventory
13 /u01/ogg/psphpp05/oracle/21.9.0.0/OPatch/opatch version
14 /u01/ogg/psphpp05/oracle/21.9.0.0/OPatch/opatch apply
```

Output

```
1 [oracle@ggpsphpp05 34958358]$ /u01/ogg/psphpp05/oracle/21.9.0.0/OPatch/opatch lsinventory
2 Oracle Interim Patch Installer version 12.2.0.1.37
```

```

3 Copyright (c) 2023, Oracle Corporation. All rights reserved.
4
5
6 Oracle Home      : /u01/ogg/psphpp05/oracle/21.9.0.0.0
7 Central Inventory : /u01/app/oraInventory
8   from           : /u01/ogg/psphpp05/oracle/21.9.0.0.0/oraInst.loc
9 OPatch version   : 12.2.0.1.37
10 OUI version      : 12.2.0.9.0
11 Log file location : /u01/ogg/psphpp05/oracle/21.9.0.0.0/cfgtoollogs/patch/patch2023-07-10_11-58-38AM_1.log
12
13 Lsinventory Output file location : /u01/ogg/psphpp05/oracle/21.9.0.0.0/cfgtoollogs/patch/lsinv/lsinventory2023-
14 07-10_11-58-38AM.txt
15 -----
16 Local Machine Information::
17 Hostname: ggpsphpp05.sbg-psp-prod.a.intuit.com
18 ARU platform id: 226
19 ARU platform description:: Linux x86-64
20
21 Installed Top-level Products (1):
22
23 Oracle GoldenGate Core 21.1.0.0.0
24 There are 1 products installed in this Oracle Home.
25
26 There are no Interim patches installed in this Oracle Home.
27
28 -----
29
30
31 OPatch succeeded.
32 [oracle@ggpsphpp05 34958358]$ /u01/ogg/psphpp05/oracle/21.9.0.0.0/OPatch/patch version
33 OPatch Version: 12.2.0.1.37
34
35 OPatch succeeded.
36 [oracle@ggpsphpp05 34958358]$ /u01/ogg/psphpp05/oracle/21.9.0.0.0/OPatch/patch apply
37 Oracle Interim Patch Installer version 12.2.0.1.37
38 Copyright (c) 2023, Oracle Corporation. All rights reserved.
39
40
41 Oracle Home      : /u01/ogg/psphpp05/oracle/21.9.0.0.0
42 Central Inventory : /u01/app/oraInventory
43   from           : /u01/ogg/psphpp05/oracle/21.9.0.0.0/oraInst.loc
44 OPatch version   : 12.2.0.1.37
45 OUI version      : 12.2.0.9.0
46 Log file location : /u01/ogg/psphpp05/oracle/21.9.0.0.0/cfgtoollogs/patch/patch2023-07-10_11-58-48AM_1.log
47
48 Verifying environment and performing prerequisite checks...
49 OPatch continues with these patches: 34958358
50
51 Do you want to proceed? [y|n]
52 y
53 User Responded with: Y
54 All checks passed.
55
56 Please shutdown Oracle instances running out of this ORACLE_HOME on the local system.
57 (Oracle Home = '/u01/ogg/psphpp05/oracle/21.9.0.0.0')
58
59
60 Is the local system ready for patching? [y|n]
61 y
62 User Responded with: Y
63 Backing up files...
64 Applying interim patch '34958358' to OH '/u01/ogg/psphpp05/oracle/21.9.0.0.0'
65
66 Patching component oracle.oggcgcore.ora21c, 21.1.0.0.0...
67 Patch 34958358 successfully applied.
68 Log file location: /u01/ogg/psphpp05/oracle/21.9.0.0.0/cfgtoollogs/patch/patch2023-07-10_11-58-48AM_1.log
69
70 OPatch succeeded.

```

• Setup Extract from Oracle

```

1 . /l/orcl
2 ogg
3 ggsci
4 info CREDENTIALSTORE
5 ALTER CREDENTIALSTORE ADD USER ggs@psphpp02.sbg-psp-prod.a.intuit.com:1521/psphpp02 password "HJadPpM5q#(8eS"
6   alias ggsources

```

```

7 dblogin useridalias ggsources
8
9 mkdir -p ./dirdat/psphnp02
10
11 ggsci
12 obey ./diroby/create_epsphnp2.oby
13 register extract epsphnp2 database

```

Output

```

1 G6SCI (ggpsphpp05.sbg-psp-prod.a.intuit.com) 1> obey ./diroby/create_epsphnp2.oby
2
3 G6SCI (ggpsphpp05.sbg-psp-prod.a.intuit.com) 2> dblogin USERIDALIAS ggsources
4
5 Successfully logged into database.
6
7 G6SCI (ggpsphpp05.sbg-psp-prod.a.intuit.com as ggs@PSPHPP02) 3> add extract epsphnp2,INTEGRATED TRANLOG, BEGIN
  NOW
8
9 Integrated Extract added.
10
11
12 G6SCI (ggpsphpp05.sbg-psp-prod.a.intuit.com as ggs@PSPHPP02) 4> add exttrail ./dirdat/psphnp02/tr extract
  epsphnp2, MEGABYTES 500
13
14 EXTTRAIL added.
15
16 G6SCI (ggpsphpp05.sbg-psp-prod.a.intuit.com as ggs@PSPHPP02) 5>
17
18 G6SCI (ggpsphpp05.sbg-psp-prod.a.intuit.com as ggs@PSPHPP02) 6> register extract epsphnp2 database
19
20 2023-07-11 10:13:49 INFO      OGG-02003 Extract group EPSPHNP2 successfully registered with database at SCN
  8399783651506.
21
22
23 G6SCI (ggpsphpp05.sbg-psp-prod.a.intuit.com as ggs@PSPHPP02) 7>

```

- Take snapshot of psp-prod-uw01 cluster After full load completes via AWS Console
- Prepare GG config files
- Endpoint setup source for and target
- Get Range Boundaries from Source
- Table wise DMS Tasks
- Table mapping
- DMS Task settings
- **Post-FullLoad steps**
 - Index creation on Target
- **Configure replication from Oracle Hash to Aurora Postgres**
- SCN details from restored snapshot database

```

1 SQL> select * from v$database_incarnation;
2
3
4
5
6
7

```

STATUS CON_ID	INCARNATION# RESETLOGS_ID	RESETLOGS_CHANGE# PRIOR_INCARNATION#	RESETLOGS FLASHBACK_DATABASE_ALLOWED	PRIOR_RESETLOGS_CHANGE# PRIOR_RES
PARENT 0	1 1133594949	629818	08-APR-23 0 NO	629237 08-APR-23
PARENT 0	2 1141923538	8399784052545	11-JUL-23 1 NO	629818 08-APR-23
CURRENT 0	3 1141924881	8399784055213	11-JUL-23 2 NO	8399784052545 11-JUL-23

```

8 SQL> select resetlogs_change# from v$database_incarnation where incarnation# = (select max(incarnation#) from
v$database_incarnation);
9
10          RESETLOGS_CHANGE#
11  -----
12          8399784055213

```

- **Configure GoldenGate on target**
- login to gg hub **ggpspag01.sbg-ppsp-prod.a.intuit.com**

```

1 cd /u01
2 mkdir -p /u01/ogg/pspag01/postgres/21.9.0.0.3
3 cd /u01/ogg/pspag01/postgres/21.9.0.0.3
4 cd /u01/software/
5 unzip p34958432_2190000GGRU_Linux-x86-64.zip
6
7 tar -xvf /u01/software/ggs_Linux_x64_PostgreSQL_64bit.tar
8 export LD_LIBRARY_PATH=/u01/ogg/pspag01/postgres/21.9.0.0.3/lib:$LD_LIBRARY_PATH
9 export PATH=/u01/ogg/pspag01/postgres/21.9.0.0.3:$PATH
10 export ODBCINI=/u01/ogg/pspag01/postgres/21.9.0.0.3/odbc.ini

```

Setup Postgres Extract and Replicat

[Setup Goldengate for Postgres \(Oracle Hash to Aurora Postgres\)](#)